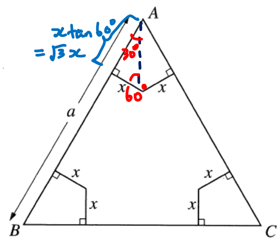
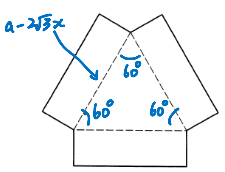
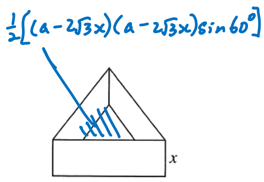
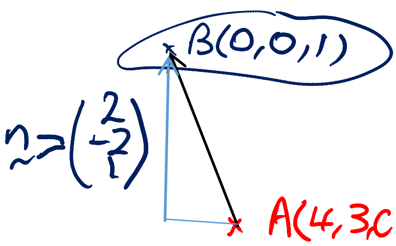
Overall Comments

- Students find these easy: **Q1, Q4(i) & (ii), Q6, Q7(ii) & (iii), Q8(i), Q9, Q10, Q12**
- Students find these challenging: **Q2, Q4(iii), Q8(iii) Q11**
- Algebraic manipulation was an area of concern.

Section A: Pure Mathematics

No.	Solutions	Comments
1i	$\mathrm{R}_g = \mathbb{R}$, $\mathrm{D}_f = \mathbb{R}\setminus\{1\}$ Since $\mathrm{R}_g \not\subset \mathrm{D}_f$, fg does not exist	
1ii	$\mathrm{gf}(x) = \mathrm{g}\left(\frac{2+x}{1-x}\right) = 1-2\left(\frac{2+x}{1-x}\right) = \frac{-3-3x}{1-x}$ Let $x = (\mathrm{gf})^{-1}(5)$ $\Rightarrow \mathrm{gf}(x) = 5$ $\Rightarrow \frac{-3-3x}{1-x} = 5$ $\Rightarrow -3-3x = 5-5x$ $\Rightarrow x = 4$	(ii) Serious errors seen e.g. work out (gf)(5) or its reciprocal.
2i	Fig. 1 Fig. 2 Fig. 3 $V = \frac{1}{2}\left[(a-2\sqrt{3}x)(a-2\sqrt{3}x)\sin 60^\circ\right](x)$ $= \frac{1}{2}(a-2\sqrt{3}x)^2\left(\frac{\sqrt{3}}{2}\right)x$ $= \frac{1}{4}x\sqrt{3}(a-2x\sqrt{3})^2$	Show how $2\sqrt{3}x$ is derived.
2ii	$\frac{\mathrm{d}V}{\mathrm{d}x} = \frac{\sqrt{3}}{4}\left[(a-2x\sqrt{3})^2 + x2(a-2x\sqrt{3})(-2\sqrt{3})\right]$ $= \frac{\sqrt{3}}{4}(a-2x\sqrt{3})\left[(a-2x\sqrt{3})-4\sqrt{3}x\right]$ $= \frac{\sqrt{3}}{4}(a-2x\sqrt{3})(a-6\sqrt{3}x)$ For stationary values of V, $\frac{\mathrm{d}V}{\mathrm{d}x} = 0$ $a-2x\sqrt{3} = 0$ or $a-6\sqrt{3}x = 0$ $a-2x\sqrt{3} = 0$ is rejected as this leads to $V = 0$ $\therefore\ a-6\sqrt{3}x = 0 \Rightarrow x = \frac{a}{6\sqrt{3}}$	Carelessness in differentiation, e.g treating a as a variable. Must give both roots to the quadratic equation before rejecting one. Give a reason for rejecting $x = \frac{a}{2\sqrt{3}}$. You are asked to **prove** that V is a maximum, hence it is easier to use the 2nd derivative test. It is not acceptable to give

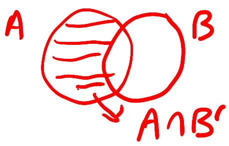
	$\dfrac{d^2V}{dx^2}=\dfrac{\sqrt{3}}{4}(-2\sqrt{3})(a-6\sqrt{3}x)+\dfrac{\sqrt{3}}{4}(a-2x\sqrt{3})(-6\sqrt{3})$ When $x=\dfrac{a}{6\sqrt{3}}$, $\dfrac{d^2V}{dx^2}=-\dfrac{9}{2}\left(a-2\dfrac{a}{6\sqrt{3}}\sqrt{3}\right)=-3a<0$ $\therefore V$ is max at $x=\dfrac{a}{6\sqrt{3}}$. $V_{\max}=\dfrac{1}{4}x\sqrt{3}(a-2x\sqrt{3})^2=\dfrac{1}{4}\left(\dfrac{a}{6\sqrt{3}}\right)\sqrt{3}\left(a-2\sqrt{3}\left(\dfrac{a}{6\sqrt{3}}\right)\right)^2$ $=\dfrac{a}{24}\left(a-\dfrac{a}{3}\right)^2=\dfrac{a^3}{54}$ cubic units	a diagram claiming to show that the gradient of $\dfrac{dV}{dx}$ was zero at the root $x=\dfrac{a}{6\sqrt{3}}$, positive before and negative afterwards, without giving any real evidence for this. You must show evidence why it changes sign from +ve to –ve, example by using a quadratic curve. Read question carefully. You must move on to find maximum volume. Remember to simplify answer to $\dfrac{a^3}{54}$
3i	$f(x)=\ln(1+2\sin x)$ $f'(x)=\dfrac{2\cos x}{1+2\sin x}$ $f''(x)=\dfrac{(1+2\sin x)(-2\sin x)-(2\cos x)(2\cos x)}{(1+2\sin x)^2}=\dfrac{-2\sin x-4}{(1+2\sin x)^2}$ $f'''(x)=\dfrac{(1+2\sin x)^2(-2\cos x)-2(1+2\sin x)(-2\sin x-4)(2\cos x)}{(1+2\sin x)^4}$ $f(0)=\ln(1+2\sin 0)=0$ $f'(0)=\dfrac{2\cos 0}{1+2\sin 0}=2$ $f''(0)=\dfrac{-2\sin 0-4}{(1+2\sin 0)^2}=-4$ $f'''(0)=\dfrac{(1+2\sin 0)^2(-2\cos 0)-2(1+2\sin 0)(-2\sin 0-4)(2\cos 0)}{(1+2\sin 0)^4}$ $=14$ $f(x)=f(0)+xf'(0)+\dfrac{x^2}{2!}f''(0)+\dfrac{x^3}{3!}f'''(0)+\ldots$ $=2x+\dfrac{x^2}{2!}(-4)+\dfrac{x^3}{3!}(14)+\ldots$ $=2x-2x^2+\dfrac{7}{3}x^3+\ldots$	carelessness in algebraic manipulation is often seen e.g. sign errors or missed out terms. Remember to simplify before further differentiating $f''(x)$.
3ii	$e^{ax}\sin nx=\left(1+ax+\dfrac{(ax)^2}{2}+\ldots\right)\left(nx-\dfrac{(nx)^3}{3!}+\ldots\right)$ $=nx+anx^2+\ldots$ $\therefore n=2,\quad an=-2\Rightarrow a=-1$	We do not need the x^3 term in the expansion of e^{ax}. Do you know why? Many did not notice that there are two ways of

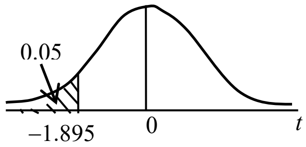
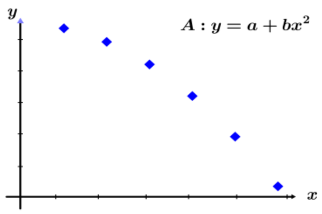
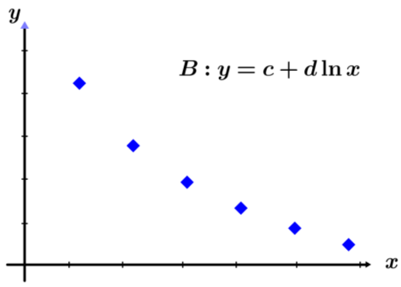
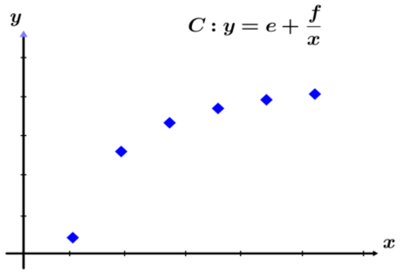
	Third non-zero term $=\frac{a^2 n}{2}x^3-\frac{n^3}{6}x^3$ $=\frac{(-1)^2(2)}{2}x^3-\frac{(2)^3}{6}x^3=-\frac{1}{3}x^3$	getting terms in x^3
4i	4(i) $\cos\theta=\frac{\begin{pmatrix}2\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}-6\\3\\2\end{pmatrix}}{3\times 7}=-\frac{16}{21}$ $\theta=139.632°$ $\therefore$ acute angle $=180°-139.632°=40.4°$	routine question but still quite a few made careless mistakes e.g. giving answer as $140°$, slip in arithmetic, subtracted $90°$ from obtuse angle e.t.c
4ii	$2x-2y+z=1 \quad -----(1)$ $-6x+3y+2z=-1 \quad -----(2)$ Using the GC to solve (1) & (2) $\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}-1/6\\-2/3\\0\end{pmatrix}+z\begin{pmatrix}7/6\\5/3\\1\end{pmatrix}$ $\therefore$ a vector equation for l is $\underset{\sim}{r}=\begin{pmatrix}-1/6\\-2/3\\0\end{pmatrix}+\lambda\begin{pmatrix}7\\10\\6\end{pmatrix}, \quad \lambda\in\mathbb{R}$.	main error being the omission of '**r** = ' in the answer
4iii	Let B (0, 0, 1) and C (0, -1, 1) be points on p_1 and p_2 respectively. $\therefore \overrightarrow{AB}=\begin{pmatrix}-4\\-3\\1-c\end{pmatrix}$ and $\overrightarrow{AC}=\begin{pmatrix}-4\\-4\\1-c\end{pmatrix}$ Distance of A from p_1 $=\left\lvert\frac{\begin{pmatrix}-4\\-3\\1-c\end{pmatrix}\cdot\begin{pmatrix}2\\-2\\1\end{pmatrix}}{3}\right\rvert=\left\lvert\frac{-1-c}{3}\right\rvert$  Distance of A from p_2 $=\left\lvert\frac{\begin{pmatrix}-4\\-4\\1-c\end{pmatrix}\cdot\begin{pmatrix}-6\\3\\2\end{pmatrix}}{7}\right\rvert=\left\lvert\frac{14-2c}{7}\right\rvert$	Need to consider the $\pm$ signs when dropping the modulus sign.

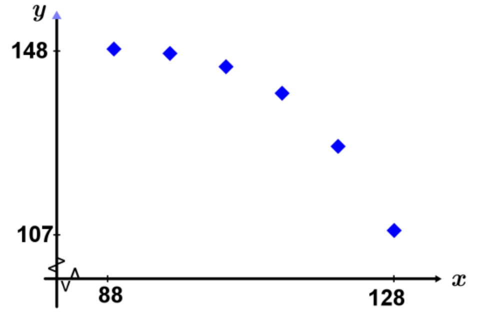
	$\therefore \left\lvert\frac{-1-c}{3}\right\rvert = \left\lvert\frac{14-2c}{7}\right\rvert$ $7+7c = 42-6c \quad \Rightarrow c = \frac{35}{13}$ or $7+7c = -42+6c \quad \Rightarrow c = -49$	

Section B: Statistics

No.	Solutions	Comments
5i	Assign the numbers 1, 2, 3, … , 100000 to the employees. Use a random number generator to obtain 90 distinct numbers, which will represent the employees invited. There may be too many from some countries and none from some countries, giving rise t o a sample that is not representative of the different countries.	Not in H2 Math syllabus
5ii	Stratified sampling. E.g. if a country has x employees, then invite $\frac{x}{100000}\times 90$ from that country. Use random sampling to pick required number of employees from each country.	Not in H2 Math syllabus Must remember to indicate how stratified sampling can be carried out. E.g. naming at least one group of strata, & mention random selection from each stratum.
6	$Y \sim \mathrm{N}\left(\mu, \sigma^2\right)$ $\mathrm{P}\left(Y < 2a\right) = 0.95 \quad \Rightarrow \mathrm{P}\left(Z < \frac{2a-\mu}{\sigma}\right) = 0.95$ $\therefore \frac{2a-\mu}{\sigma} = 1.6449 \quad ----(1)$ $\mathrm{P}\left(Y < a\right) = 0.25 \quad \Rightarrow \mathrm{P}\left(Z < \frac{a-\mu}{\sigma}\right) = 0.25$ $\therefore \frac{a-\mu}{\sigma} = -0.67449 \quad ----(2)$ $\frac{(1)}{(2)}: \quad \frac{2a-\mu}{a-\mu} = \frac{1.6449}{-0.67449}$ $\Rightarrow 2a-\mu = -2.4387a + 2.4387\mu$ $\Rightarrow 4.4387a = 3.4387\mu$ $\therefore \mu = 1.29a$	
7i	For F to be well modelled by a binomial distribution, the probability that a packet containing a free gift remains constant and that whether a packet contains a free gift or not is independent of whether the other packets contain a free gift or not.	Only a minority were successful in giving both assumptions. Wrong statements given includes (1) a packet either

		contains a gift or it does not ☹ OR (2) probability of a packet containing a gift is independent of other packets ☹
7ii	$F \sim \mathrm{B}(20, 0.05)$ $\mathrm{P}(F=1) = 0.377$	
7iii	$F \sim \mathrm{B}(60, 0.05)$ Since n is large, $np = 3 < 5$, $F \sim \mathrm{Po}(3)$ $\therefore \mathrm{P}(F \geq 5) = 1 - \mathrm{P}(F \leq 4)$ $= 0.185$	Not in H2 math syllabus Errors include using normal distribution as the approximation OR finding $\mathrm{P}(F > 5)$ instead.
8i	8(i) $\mathrm{P}(B \cap A') = \mathrm{P}(B \mid A') \times \mathrm{P}(A')$ $= (0.8)(1 - 0.7)$ $= 0.24$	
8ii	Alternative (ii) $\mathrm{P}(A' \cap B') + \mathrm{P}(B \cap A') = \mathrm{P}(A')$ $\quad P(A' \cap B')$ $\mathrm{P}(A' \cap B') = \mathrm{P}(A') - \mathrm{P}(B \cap A') = 1 - P(A) - P(B \cap A')$ $= 0.3 - 0.24$ $= 0.06$	
8iii	$\mathrm{P}(A \mid B') = 0.88$ $\qquad P(A \cap B') = P(A) - P(A \cap B)$ $\dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = 0.88$  $\Rightarrow \dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(A \cap B') + \mathrm{P}(A' \cap B')} = 0.88$ $\Rightarrow \mathrm{P}(A \cap B') = 0.88\left[\mathrm{P}(A \cap B') + 0.06\right]$ $\Rightarrow \mathrm{P}(A \cap B') = \dfrac{0.88 \times 0.06}{0.12} = 0.44$ $\Rightarrow \mathrm{P}(A \cap B) = P(A) - \mathrm{P}(A \cap B') = 0.26$	This is tough for many students. You need to find $\mathrm{P}(A \cap B')$ first and use it to find $\mathrm{P}(A \cap B)$.
9i	Let X km be the distance travelled per litre of fuel by a car and μ km be the population mean of X. From the GC, Unbiased estimate of population mean = $\bar{x} = 12.8$ Unbiased estimate of population variance $= s^2 = 2.3058 \approx 2.31$	Key in the data given and read off S_x from the GC. Remember to square this value to get the unbiased estimate of the population variance.

9ii	$H_0: \mu = 13.8$ $H_1: \mu < 13.8$ Level of significance: 5% Fill in this value Test statistic: $T = \dfrac{\overline{X} - 13.8}{\frac{S}{\sqrt{n}}}$  Computation: $\bar{x} = 12.8$, $s^2 = 2.3058$, $n = 8$ $p-\text{value} = 0.0524$ Conclusion: Since p-value = 0.0525 > 0.05, we do not reject H_0. There is insufficient evidence at 5% significance level that the figures quoted by car manufacturers for distances travelled per litre of fuel are too high Assumption: The given set of data from the eight readers constitutes a random sample from the population of distances travelled per litre of fuel for cars of the specified model. **Alternatively**, The responses given by the eight readers are independent of one another.	Not in H2 Math syllabus Do not say "the distances travelled per litre are normally distributed" since that is already given to be so in the question.
10i	  	Follow the instructions e.g. include 6 points, use a separate diagram for each part, all x- and y-values positive. Do note that because x-intervals are constant, so y-intervals should either increase for A and decrease for B and C.

10ii		
10iii	Model A is the most appropriate since y values decreases by increasing amount as x increases, which fits the curvature seen in the scatter diagram. $r = -0.939$	Almost all identified (A) although not all gave a reason.
10iv	From the GC, $y = 189.75 - 0.0046198x^2$ When $x = 110$, $y = 189.75 - 0.0046198(110)^2 \approx 134$	
11i	P(three different letters & 2 different digits in the code) $= \dfrac{\left({}^{26}P_3\right)\left({}^{9}P_2\right)}{\left(26^3\right)\left(9^2\right)}$ $\dfrac{26 \times 25 \times 24 \times 9 \times 8}{26^3 \times 9^2}$ $= \dfrac{400}{507} \approx 0.789$	Many candidates found this question challenging. Students had difficulties working out which numbers should be multiplied together; unnecessary factors of 3 and 2 were seen quite often
11ii	P(the second digit higher than the first digit in the code) $= \dfrac{\left({}^{9}P_2\right)}{(2)\left(9^2\right)}$ $= \dfrac{4}{9} \approx 0.444$	Alternatively, Consider the first digit as 1, 2, 3, …, 8. Then the 2 nd digit has 8, 7, 6, … 1 options respectively. Pick any 2 digits and arrange. Half of those arrangements would be such that the 2 nd digit is higher than the first.
11iii	P(exactly two letters the same) $= \dfrac{\left({}^{26}C_2\right)\left({}^{2}C_1\right)\frac{3!}{2!}}{26^3} = \dfrac{75}{676}$ P(exactly two digits the same) $= \dfrac{9}{9^2} = \dfrac{1}{9}$ P(exactly two letters the same and two digits the same)	A lot of students did not consider all the cases in their solutions.

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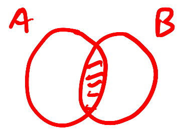
$=\dfrac{75}{676}\times\dfrac{1}{9}=\dfrac{75}{6084}$

Required probability

$=\dfrac{75}{676}+\dfrac{1}{9}-2\left(\dfrac{75}{6084}\right)=\dfrac{1201}{6084}$



OR

P(exactly two letters the same or two digits the same, but not both)
= P(two letters the same & two digits different) + P(all letters different & two digits the same) + P(all letters the same & two digits the same)

$=\dfrac{\left({}^{26}C_2\left({}^{2}C_1\right)\dfrac{3!}{2!}\right)\left({}^{9}P_2\right)+\left({}^{26}P_3\right)(9)+(26)(9)}{\left(26^3\right)\left(9^2\right)}$

$=0.197403$

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P(exactly one vowel [A, E, I, O or U] and exactly one even digit)

$=\dfrac{\left({}^{4}C_1\times{}^{5}C_1\times 2!\right)\left[\left({}^{5}C_1\right)\left({}^{21}C_2\right)(3!)+\left({}^{5}C_1\right)\left({}^{21}C_1\right)(3)\right]}{\left(26^3\right)\left(9^2\right)}$

$=0.186$

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9 < 4 even, 5 odd

26 < 5 vowels, 21 Consonants

Some students forgot the factor 3 to take account of the different positions of the vowel and the factor 2 to account for the different positions of the even digit. Others used 3! instead.

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Two conditions:

(1) Employees absent through illness are independent of each other,
(2) The average rate of employees absent through illness is a constant OR the mean number of employees absent through illness in a given time interval is proportional to the size of the interval

Condition (1): The average rate of employees absent through illness cannot be a constant because more people especially those with weaker health immunities are likely to fall ill easily during cold seasons or drastic changes in weather conditions.

Condition (2): Employees absent through illness are not independent of each other as infectious illnesses such as common flu are easily passed on from employee to employee working in the same department.

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12ii	Let X be the number of absences through illness in n days for the Administration Department $X \sim \mathrm{Po}(1.2n)$ $\mathrm{P}(X=0)<0.01$ $\Rightarrow \mathrm{e}^{-1.2n}<0.01$ $\Rightarrow -1.2n<\ln(0.01)$ $\Rightarrow n>\dfrac{\ln(0.01)}{-1.2}$ $\Rightarrow n>3.84$ Smallest number of days required is 4.	Not in H2 Math syllabus Alternatively, n | $\mathrm{P}(X=0)$ 3 | 0.02732 (> 0.01) 4 | 0.00823 (< 0.01) Smallest number of days required is 4.
12iii	Let X and Y be the number of days of absence in 5 days for the Administration Department and Manufacturing Department respectively $X \sim \mathrm{Po}(1.2\times 5), \quad Y \sim \mathrm{Po}(2.7\times 5)$ $X+Y \sim \mathrm{Po}(6+13.5)$ i.e. $X+Y \sim \mathrm{Po}(19.5)$ $\mathrm{P}(X+Y>20)=1-\mathrm{P}(X+Y\le 20)$ $=0.397$	Not in H2 Math syllabus
12iv	Let X and Y be the number of days of absence in 60 days for the Administration Department and Manufacturing Department respectively $X \sim \mathrm{Po}(1.2\times 60), \quad Y \sim \mathrm{Po}(2.7\times 60)$ $X+Y \sim \mathrm{Po}(72+162)$ i.e. $X+Y \sim \mathrm{Po}(234)$ Since $\lambda=234>10$, $X+Y \sim \mathrm{N}(234, 234)$ approximately. $\mathrm{P}(200\le X+Y\le 250)=\mathrm{P}(199.5\le X+Y\le 250.5)=0.848$	Not in H2 Math syllabus